

Macro HW2 — Notes

ECON*6020 Macroeconomic Theory I, Assignment 2 (Arrow learning-by-investing)

Model.

$$\dot{k} = s(w + rk) - \delta k,$$
$$Y_i = (AL_i)^{1-\alpha} K_i^\alpha, \quad A = a \left(\frac{1}{N} \sum_{i=1}^N K_i \right)^b.$$

L identical consumers each supply one unit of labour and rent capital to N firms. Each firm has constant returns in its own K_i, L_i , but technology A rises with the *average* capital stock. Firms take A as given, so capital accumulation creates an externality (Arrow's learning-by-investing).

(a) Market clearing in the factor markets

In each factor market, the total quantity demanded by firms equals the total quantity supplied by all consumers:

$$L = \sum_{i=1}^N L_i, \quad K = k \cdot L = \sum_{i=1}^N K_i.$$

Labour: every consumer supplies one unit, so the firms together hire all L workers. Capital: the capital consumers own, kL , is all rented out to the firms.

(b) Real wage and rental rate

Derive r and w as functions of the model parameters and the average capital-labour ratio.

Competitive firms pay each factor its marginal product, taking A as given. Because all firms face the same w, r and A and have constant returns, they all choose the same ratio $K_i/L_i = K/L = k$. We then substitute $A = a(K/N)^b$ and $K = Lk$.

Wage.

$$\begin{aligned} w &= \frac{\partial (AL_i)^{1-\alpha} K_i^\alpha}{\partial L_i} = (1-\alpha) A^{1-\alpha} L_i^{-\alpha} K_i^\alpha \\ &= (1-\alpha) \left[a \left(\frac{K}{N} \right)^b \right]^{1-\alpha} \left(\frac{K_i}{L_i} \right)^\alpha = (1-\alpha) a^{1-\alpha} \left(\frac{K}{N} \right)^{(1-\alpha)b} k^\alpha \\ &= (1-\alpha) a^{1-\alpha} \left(\frac{L \cdot k}{N} \right)^{(1-\alpha)b} k^\alpha \\ &= (1-\alpha) a^{1-\alpha} \left(\frac{L}{N} \right)^{(1-\alpha)b} k^{(1-\alpha)b + \alpha}. \end{aligned}$$

Rental rate. With $Y_i = (AL_i)^{1-\alpha} K_i^\alpha$ and $K = L \cdot k$:

$$\begin{aligned} r &= \frac{\partial Y_i}{\partial K_i} = \alpha A^{1-\alpha} L_i^{1-\alpha} K_i^{\alpha-1} = \alpha \left\{ a \left[\left(\frac{L \cdot k}{N} \right) \right]^b \right\}^{1-\alpha} k^{\alpha-1} \\ &= \alpha a^{1-\alpha} \left(\frac{L}{N} \right)^{b(1-\alpha)} k^{b(1-\alpha) + (\alpha-1)} = \alpha a^{1-\alpha} \left(\frac{L}{N} \right)^{b(1-\alpha)} k^{(1-\alpha)(b-1)}. \end{aligned}$$

(c) Increasing returns to scale for $b > 0$

Idea: add up output across firms to get aggregate output Y as a function of K and L , then scale both inputs by z and compare the result with zY .

$$\begin{aligned}
Y_i &= (AL_i)^{1-\alpha} K_i^\alpha, \\
Y &= \sum_{i=1}^N (AL_i)^{1-\alpha} K_i^\alpha = \sum_{i=1}^N A^{1-\alpha} L_i \cdot L_i^{-\alpha} K_i^\alpha = \sum_{i=1}^N A^{1-\alpha} L_i \left(\frac{K_i}{L_i}\right)^\alpha \\
&= \sum_{i=1}^N A^{1-\alpha} k^\alpha L_i = A^{1-\alpha} k^\alpha \sum_{i=1}^N L_i = A^{1-\alpha} k^\alpha L \\
&= a^{1-\alpha} \left(\frac{\sum_{i=1}^N K_i}{N}\right)^{b(1-\alpha)} k^\alpha L = a^{1-\alpha} \left(\frac{K}{N}\right)^{b(1-\alpha)} k^\alpha L \\
&= a^{1-\alpha} \left(\frac{K}{N}\right)^{b(1-\alpha)} L \left(\frac{K}{L}\right)^\alpha.
\end{aligned}$$

If we multiply both inputs K and L by z , we get

$$\begin{aligned}
&a^{1-\alpha} \left(\frac{zK}{N}\right)^{b(1-\alpha)} zL \left(\frac{zK}{zL}\right)^\alpha \\
&= z^{b-b\alpha+1} a^{1-\alpha} \left(\frac{K}{N}\right)^{b(1-\alpha)} L \left(\frac{K}{L}\right)^\alpha = z^{1+b(1-\alpha)} Y.
\end{aligned}$$

Since $0 < \alpha < 1$ (and $b > 0$),

$$0 < b\alpha < b \implies b - b\alpha > 0 \implies b - b\alpha + 1 > 1.$$

Therefore, for $z > 1$, output rises by more than the factor z by which K and L are increased: the economy as a whole has increasing returns to scale in capital and labour.

Intuition: each firm alone has constant returns, but when all firms raise capital together the average capital stock rises, which raises A for everyone. That spillover gives the extra $z^{b(1-\alpha)}$. If $b = 0$ there is no spillover and we are back to constant returns.

(d) Transition dynamics to the steady state

Plug w and r from (b) into the budget constraint $\dot{k} = s(w + rk) - \delta k$:

$$\begin{aligned}
w + rk &= (1 - \alpha)a^{1-\alpha} \left(\frac{L}{N}\right)^{(1-\alpha)b} k^{(1-\alpha)b+\alpha} + \alpha a^{1-\alpha} \left(\frac{L}{N}\right)^{b(1-\alpha)} k^{(1-\alpha)(b-1)+1} \\
&= a^{1-\alpha} \left(\frac{L}{N}\right)^{b(1-\alpha)} \left[(1 - \alpha)k^{(1-\alpha)b+\alpha} + \alpha k^{(1-\alpha)b+\alpha} \right] \\
&= a^{1-\alpha} \left(\frac{L}{N}\right)^{b(1-\alpha)} k^{(1-\alpha)b+\alpha} = y.
\end{aligned}$$

This matches output per worker from (c): $y = Y/L = a^{1-\alpha} \left(\frac{kL}{N}\right)^{b(1-\alpha)} k^\alpha = a^{1-\alpha} \left(\frac{L}{N}\right)^{b(1-\alpha)} k^{b(1-\alpha)+\alpha}$. So income per worker is all spent or saved, and

$$\dot{k} = sy - \delta k = sa^{1-\alpha} \left(\frac{L}{N}\right)^{b(1-\alpha)} k^{b(1-\alpha)+\alpha} - \delta k.$$

Dividing by k gives the growth rate of capital per worker:

$$\gamma_k \equiv \frac{\dot{k}}{k} = sa^{1-\alpha} \left(\frac{L}{N} \right)^{b(1-\alpha)} k^{(1-\alpha)(b-1)} - \delta.$$

Everything depends on the sign of the exponent $(1-\alpha)(b-1)$. Since $y \propto k^{b(1-\alpha)+\alpha}$, income per worker grows at $\gamma_y = [b(1-\alpha) + \alpha] \gamma_k$, so it follows the same pattern.

(i) $b < 1$: stable steady state. The exponent is negative, so the growth rate falls as k rises. Setting $\gamma_k = 0$,

$$k^* = \left[\frac{sa^{1-\alpha} (L/N)^{b(1-\alpha)}}{\delta} \right]^{\frac{1}{(1-\alpha)(1-b)}}.$$

If $k < k^*$ then $\gamma_k > 0$, and growth slows as k approaches k^* ; if $k > k^*$ then k shrinks. The economy converges to k^* just as in the Solow model, and long-run growth is zero.

(ii) $b > 1$: unstable steady state. The exponent is positive, so the growth rate *rises* with k . The same k^* exists but is unstable. If $k > k^*$, growth is positive and keeps speeding up (explosive growth). If $k < k^*$, k shrinks faster and faster toward zero. There is no convergence.

(iii) $b = 1$: no transition dynamics. The exponent is zero, so γ_k does not depend on k . The economy grows at one constant rate from the start (an AK model); see (e).

(e) Steady-state growth rate for $b = 1$

With $b = 1$, output per worker is linear in k : $y = a^{1-\alpha} (L/N)^{1-\alpha} k$. So

$$\gamma = \frac{\dot{k}}{k} = \frac{\dot{y}}{y} = sa^{1-\alpha} \left(\frac{L}{N} \right)^{1-\alpha} - \delta.$$

This is the steady-state growth rate of capital and income per worker. With no population growth it is also the growth rate of total output Y . Growth is positive if $sa^{1-\alpha} (L/N)^{1-\alpha} > \delta$, and it rises with the savings rate s . It also rises with L/N , the number of workers per firm (a scale effect).